

**MARK SCHEME for the October/November 2009 question paper
for the guidance of teachers**

9709 MATHEMATICS

9709/32

Paper 32, maximum raw mark 75

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

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Page 2	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9709	32

Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more “method” steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously “correct” answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9709	32

The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no “follow through” from a previous error is allowed)
CWO	Correct Working Only – often written by a ‘fortuitous’ answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1	A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become “follow through ✓” marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR –2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA –1	This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9709	32

- 1 Use law of the logarithm of a product or quotient and remove logarithms M1
Obtain quadratic equation $x^2 - 5x + 5 = 0$, or equivalent A1
Solve 3-term quadratic obtaining 1 or 2 roots A1
Obtain answers 1.38 and 3.62 A1 [4]
- 2 (i) Evaluate, or consider the sign of, $x^3 - 8x - 13$ for two integer values of x , or equivalent M1
Conclude $x = 3$ and $x = 4$ with no errors seen A1 [2]
- (ii) Use the iterative formula correctly at least once M1
Obtain final answer 3.43 A1
Show sufficient iterations to at least 4 d.p. to justify its accuracy to 2 d.p., or show there is a sign change in the interval (3.425, 3.435) A1 [3]
- 3 (i) State $2xy + x^2 \frac{dy}{dx}$ as derivative of x^2y B1
State $3y^2 \frac{dy}{dx}$ as derivative of y^3 B1
Equate derivative of LHS to zero and solve for $\frac{dy}{dx}$ M1
Obtain answer $\frac{3x^2 - 2xy}{x^2 + 3y^2}$, or equivalent A1 [4]
- (ii) Find gradient of tangent at (2, 1) and form equation of tangent M1
Obtain answer $8x - 7y - 9 = 0$, or equivalent A1✓ [2]
- 4 Use $\tan(A \pm B)$ formula and obtain an equation in $\tan \alpha$ and $\tan \beta$ M1*
Substitute throughout for $\tan \alpha$ or for $\tan \beta$ M1(dep*)
Obtain $2 \tan^2 \beta + \tan \beta - 1 = 0$ or $\tan^2 \alpha + \tan \alpha - 2 = 0$, or equivalent A1
Solve a 3-term quadratic and find an angle M1
Obtain answer $\alpha = 45^\circ, \beta = 26.6^\circ$ A1
Obtain answer $\alpha = 116.6^\circ, \beta = 135^\circ$ A1 [6]
[Treat answers given in radians as a misread. Ignore answers outside the given range.]
[SR: Two correct values of α (or β) score A1; then A1 for both correct α, β pairs]
- 5 (i) Substitute $x = -2$, equate to zero and state a correct equation, e.g. $-16 + 4a - 2b - 4 = 0$ B1
Differentiate $p(x)$, substitute $x = -2$ and equate to zero M1
Obtain a correct equation, e.g. $24 - 4a + b = 0$ A1
Solve for a or for b M1
Obtain $a = 7$ and $b = 4$ A1 [5]
- (ii) EITHER: State or imply $(x + 2)^2$ is a factor B1
Attempt division by $(x + 2)^2$ reaching a quotient $2x + k$ or use inspection with unknown factor $cx + d$ reaching $c = 2$ or $d = -1$ M1
Obtain factorisation $(x + 2)^2(2x - 1)$ A1
OR: Attempt division by $(x + 2)$ M1
Obtain quadratic factor $2x^2 + 3x - 2$ A1
Obtain factorisation $(x + 2)(x + 2)(2x - 1)$ A1 [3]
[The M1 is earned if division reaches a partial quotient of $2x^2 + kx$, or if inspection has an unknown factor of $2x^2 + ex + f$ and an equation in e and/or f , or if two coefficients with the correct moduli are stated without working.]

Page 5	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9709	32

- 6 (i) State or imply $\frac{dx}{d\theta} = 2\sec^2 \theta$ or $dx = 2 \sec^2 \theta d\theta$ B1
Substitute for x and dx throughout M1
Obtain any correct form in terms of θ A1
Obtain the given form correctly (including the limits) A1 [4]
- (ii) Use $\cos 2A$ formula, replacing integrand by $a + b \cos 2\theta$, where $ab \neq 0$ M1*
Integrate and obtain $\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta$ A1
Use limits $\theta = 0$ and $\theta = \frac{1}{4}\pi$ M1(dep*)
Obtain answer $\frac{1}{8}(\pi + 2)$, or exact equivalent A1 [4]
- 7 (i) (a) State that $u + v$ is equal to $1 + 2i$ B1 [1]
- (b) EITHER: Multiply numerator and denominator of u/v by $3 - i$, or equivalent M1
Simplify numerator to $-5 + 5i$, or denominator to 10 A1
Obtain answer $-\frac{1}{2} + \frac{1}{2}i$, or equivalent A1
OR1: Obtain two equations in x and y and solve for x or for y M1
Obtain $x = -\frac{1}{2}$ or $y = \frac{1}{2}$ A1
Obtain answer $-\frac{1}{2} + \frac{1}{2}i$, or equivalent A1
OR2: Using the correct processes express u/v in polar form M1
Obtain $x = -\frac{1}{2}$ or $y = \frac{1}{2}$ correctly A1
Obtain answer $-\frac{1}{2} + \frac{1}{2}i$, or equivalent A1 [3]
- (ii) State that the argument of u/v is $\frac{3}{4}\pi$ (2.36 radians or 135°) B1✓ [1]
- (iii) EITHER: Use facts that angle $AOB = \arg u - \arg v$ and $\arg u - \arg v = \arg(u/v)$ M1
Obtain given answer A1
OR1: Obtain $\tan \hat{AOB}$ from gradients of OA and OB and the $\tan(A \pm B)$ formula M1
Obtain given answer A1
OR2: Obtain $\cos \hat{AOB}$ by using the cosine formula or scalar product M1
Obtain given answer A1 [2]
- (iv) State $OA = BC$ B1
State OA is parallel to BC B1 [2]
- 8 (i) State or imply partial fractions are of the form $\frac{A}{1-x} + \frac{Bx+C}{2+x^2}$ B1
Use a relevant method to determine a constant M1
Obtain $A = \frac{2}{3}$, $B = \frac{2}{3}$ and $C = \frac{1}{3}$ A1 + A1 + A1 [5]

Page 6	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9709	32

- (ii) Use correct method to find first two terms of the expansion of $(1-x)^{-1}$, $(2+x^2)^{-1}$ or $(1 + \frac{1}{2}x^2)^{-1}$ M1
Obtain complete unsimplified expansions up to x^2 of each partial fraction e.g. $\frac{2}{3}(1+x+x^2)$
and $\frac{1}{2}(\frac{2}{3}x - \frac{1}{3})(1 - \frac{1}{2}x^2)$ A1✓ + A1✓
Carry out multiplication of $(2+x^2)^{-1}$ by $(\frac{2}{3}x - \frac{1}{3})$, or equivalent, provided $BC \neq 0$ M1
Obtain answer $\frac{1}{2} + x + \frac{3}{4}x^2$ A1 [5]
[Symbolic binomial coefficients are not sufficient for the first M1. The f.t. is on A, B, C .]
[If B or C omitted from the form of fractions, give B0M1A0A0A0 in (i); M1A1✓A1✓ in (ii), max 4/10]
[In the case of an attempt to expand $(1+x)(1-x)^{-1}(2+x^2)^{-1}$, give M1A1A1 for the expansions, M1 for multiplying out fully, and A1 for the final answer.]
[Allow Maclaurin, giving M1A1✓A1✓ for differentiating and obtaining $f(0) = \frac{1}{2}$ and $f'(0) = 1$, A1✓ for $f''(0) = \frac{3}{2}$, and A1 for the final answer (the f.t. is on A, B, C if used).]

- 9 (i) Separate variables correctly B1
Integrate and obtain term $\ln(\theta - A)$, or equivalent B1
Integrate and obtain term $-kt$, or equivalent B1
Use $\theta = 4A$, $t = 0$ to determine a constant, or as limits M1
Obtain correct answer in any form, e.g. $\ln(\theta - A) = -kt + \ln 3A$, with no errors seen A1 [5]
(ii) Substitute $\theta = 3A$, $t = 1$ and justify the given statement B1 [1]
(iii) Substitute $t = 2$ and solve for θ in terms of A M1
Remove logarithms M1
Obtain answer $\theta = \frac{7}{3}A$, or equivalent, with no errors seen A1 [3]
[The M marks are only available if the solution to part (i) contains terms $a\ln(\theta - A)$ and bt .]
- 10 (i) Substitute coordinates $(1, 4, 2)$ in $2x - 3y + 6z = d$ M1
Obtain plane equation $2x - 3y + 6z = 2$, or equivalent A1 [2]
(ii) EITHER: Attempt to use plane perpendicular formula to find perpendicular from $(1, 4, 2)$ to p M1
Obtain a correct unsimplified expression, e.g. $\frac{|2 - 3(4) + 6(2) - 16|}{\sqrt{(2^2 + (-3)^2 + 6^2)}}$ A1
Obtain answer 2 A1
OR1: State or imply perpendicular from O to p is $\frac{16}{7}$, or from O to q is $\frac{2}{7}$, or equivalent B1
Find difference in perpendiculars M1
Obtain answer 2 A1
OR2: Obtain correct parameter value, or position vector or coordinates of foot of perpendicular from $(1, 4, 2)$ to p ($\mu = \pm \frac{2}{7}$; $(\frac{11}{7}, \frac{22}{7}, \frac{26}{7})$) B1
Calculate the length of the perpendicular M1
Obtain answer 2 A1

Page 7	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A/AS LEVEL – October/November 2009	9709	32

- OR3:* Carry out correct method for finding the projection onto a normal vector of a line segment joining a point on p , e.g. (8, 0, 0) and a point on q , e.g. (1, 4, 2) M1
- Obtain a correct unsimplified expression, e.g. $\frac{|2(8-1) - 3(-4) + 6(-2)|}{\sqrt{(2^2 + (-3)^2 + 6^2)}}$ A1
- Obtain answer 2 A1 [3]
- (iii) *EITHER:* Calling the direction vector $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, use scalar product to obtain a relevant equation in a , b and c M1*
- Obtain two correct equations, e.g. $2a - 3b + 6c = 0$, $a - 2b + 2c = 0$ A1
- Solve for one ratio, e.g. $a : b$ M1(dep*)
- Obtain $a : b : c = 6 : 2 : -1$, or equivalent A1
- State answer $\mathbf{r} = \lambda(6\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ or equivalent A1✓
- OR:* Attempt to calculate vector product of two normals, e.g. $(\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}) \times (2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k})$ M2
- Obtain two correct components A1
- Obtain $-6\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, or equivalent A1
- State answer $\mathbf{r} = \lambda(-6\mathbf{i} - 2\mathbf{j} + \mathbf{k})$, or equivalent A1✓ [5]